PRATYUSH DHUNGANA

Forest Biometrician & Geospatial Data Scientist | Forest Carbon | GIS Automation

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PROFESSIONAL SUMMARY

Data Scientist with a foundation in forestry, forest biometrics, statistics, and geospatial analysis, with professional experience applying these disciplines in analyst roles. My work has brought together field observations, forest and vegetation data, remote sensing, LiDAR, spatial analysis, and statistical methods to understand real-world systems and solve practical problems. I am comfortable moving from an open question or imperfect dataset to a defensible analysis, whether that means investigating patterns, evaluating data quality and uncertainty, developing a spatial workflow, automating a process, or producing information that supports a decision. I bring the perspective of a scientist with the practical mindset of an analyst: understand the problem, examine the evidence, and build something useful from it.

AREAS OF EXPERTISE

Forest Management & Timber Economics: Forest management planning · Silviculture · Stand dynamics · Growth & yield · Harvest scheduling & optimization (linear programming) · Timber valuation · DCF / NPV / LEV · 10-year management plans

Biometrics, Estimation & Modeling: Forest inventory & mensuration · Height–diameter modeling · Sampling design · Small area estimation · Aboveground biomass & carbon estimation · Statistical & machine-learning models · Model validation · Uncertainty assessment

Geospatial & Remote Sensing: GIS · High-density LiDAR · Satellite & aerial imagery · Google Earth Engine · Multi-temporal change analysis · Enterprise GIS (ArcGIS Portal, feature & REST services) · Cartography

Programming, Automation & Data: Python · ArcPy toolbox development · R (lidR, terra, tidyverse) · SQL · geoprocessing automation · FastAPI · R Shiny · JavaScript (React, D3) · PostGIS / SQLite / R-tree · GDAL · PDAL · Git

PROFESSIONAL EXPERIENCE

Associate Geospatial Analyst **Dec 2025 – Sep 2026**

The Spear Group, on behalf of Florida Power & Light · West Palm Beach, FL

- Developed automated LiDAR-to-raster workflows using Python and ArcPy to quantify existing vegetation volume and projected removal volume across electric-utility corridors, supporting vegetation-management planning.
- Produced spatial distribution maps of vegetation volume and removal requirements, translating LiDAR-derived measurements into location-specific information for operational prioritization.
- Engineered a full-stack LiDAR data retrieval application using Python, R Shiny, SQLite, and R-tree spatial indexing to automate area-of-interest tile discovery, retrieval, and batch export — reducing multi-day data-staging workflows to minutes.
- Automated processing of 14,000+ high-resolution LiDAR tiles for raster generation, feature extraction, and large-scale utility vegetation analysis.
- Co-led a 38,616+ mile positional-accuracy assessment of electric-utility network data using LiDAR-derived ground truth, spatial statistics, and multi-ring buffer analysis across transmission, feeder, and lateral assets.
- Developed spatial analyses for vegetation-encroachment forecasting, drone placement, and road reachability by integrating LiDAR, remote sensing, spatial statistics, and machine learning.
- Built and documented 10 custom ArcPy geoprocessing tools, including end-to-end user tutorials and technical documentation.
- Designed geospatial schemas and published feature services through an ArcGIS Enterprise Portal environment; used ArcGIS REST services to access enterprise GIS resources and service-based data workflows.
- Supported enterprise application operations for GEVI (GE Visual Intelligence) — a platform for planning and scoping arborist and vegetation-management work — as assistant to the project manager: coordinated user roles and permissions, submitted and tracked vendor support tickets, attended recurring data-ingestion, user-feedback, and release-planning meetings, and contributed to release notes and user status updates.

Graduate Teaching & Research Assistant **Jan 2024 – Dec 2025**

Forest Biometrics Lab, Mississippi State University · Starkville, MS

- Conducted graduate research developing small area estimation methods for county-level aboveground forest biomass, combining FIA forest-inventory data with remote-sensing-derived auxiliary information — the statistical foundation of forest carbon-stock quantification.
- Developed and evaluated statistical and machine-learning models to estimate forest attributes across geographic domains, assessing performance with quantitative accuracy and uncertainty measures.
- Built reproducible Python and R workflows automating preprocessing, quality control, statistical analysis, and model evaluation for large forest-inventory and remote-sensing datasets.
- Co-instructed undergraduate laboratories in forest inventory, sampling design, stand dynamics, statistics, regression, and GIS.

Research Assistant **Sep 2023 – Dec 2023**

NAAMII · Lalitpur, Nepal

- Led a three-person team through ground-truth collection, geospatial data preparation, image labeling, and model development and evaluation for automated tree-species classification from high-resolution satellite imagery using YOLOv5.

Forestry Consultant **Apr 2023 – Jun 2023**

Independent · Nepal

- Conducted preliminary resource assessments with forest officials and local stakeholders, including participatory resource mapping, to identify forest resources, management priorities, and operational considerations across 30+ community forests.
- Conducted forest-boundary surveys, developed field sampling designs, and collected inventory data to quantify standing forest resources.
- Analyzed inventory and spatial information to support management recommendations and standing-stock revenue estimates using established timber prices.
- Prepared formal 10-year forest management plans and submitted reports to the Division Forest Office; plans were reviewed and approved for implementation as official management plans.

GIS / Forestry Intern **Feb 2022 – Jun 2022**

Forest Research and Training Center / Division Forest Office · Myagdi, Nepal

- Designed a geodatabase for 30+ community forests, integrating scanned maps, field surveys, digitized features, and aerial photography.
- Developed and compared machine-learning models for land-cover classification from remotely sensed data.
- Led an 8-person field team through a 20-day bat acoustic monitoring and spatial-data collection campaign, and ran a GIS workshop for 300+ high-school students on burned-area detection and mapping from satellite imagery.

Remote Intern **Jan 2021 – Jul 2021**

Friends of Nature (FON Nepal)

- Designed questionnaires, analyzed survey data, and prepared conservation reports on wildlife perception and conservation status.
- Organized conservation outreach for 150+ high-school students and developed educational materials for local communities.

SELECTED PROJECTS

- **FIA Tree Height Codes (3-part analysis)** — independent data-validation series on USDA FIA tree heights using PostgreSQL/PostGIS, SQLite, and Python: (1) trained height–diameter models on 11.8M FIA trees and evaluated ~200,000 crew-reconstructed heights on broken-top trees, diagnosing a spurious stump-height artifact by simulation; (2) benchmarked FIA's undocumented height model against each tree's remeasurement history; (3) compared design-based live-tree carbon density across 35 ecodivisions (Welch's ANOVA, Kruskal–Wallis, post-hoc tests), tracing high within-division variance to FIA's sample design. Public write-ups and code.
- **3D Forest Height Studio** (2026) — built an open-source R Shiny / three.js app that turns ETH canopy-height data into interactive 3D forest maps for any region, with PNG, web-page, GeoTIFF, and R-script export; github.com/pratyush-dh/3d-forest-height-studio.
- **Rainfall Relief & Pulse** (2026) — built an interactive map (three.js, MapLibre GL, Python/rasterio) that draws PRISM US annual precipitation as terrain height beside real elevation, with a 30-year animation, user-uploaded yearly rasters, and per-year map and GIF export; added a drought-index dashboard and a 1-acre hexagon map.
- **US Wood-Products Mills Map** (2026) — compiled about 196 pulp, paper, sawmill, plywood, OSB, and MDF mills from public sources into a D3 time-series map (1860 to present) with type filters and CSV loading.

- **Carbon Data Atlas** (2026) — built and deployed an open web atlas (React, D3) cataloging 16 satellite, model, inventory, and ground-based CO_2, CH_4, and N_2O datasets, with coverage maps and metadata export; pratyush-dh.github.io/carbon-data-atlas.
- **Sight: Pokhara Viewpoints** (2026) — 360° line-of-sight study of nine Himalayan peaks from seven viewpoints (Python).
- **County-Level Forest Biomass Estimation** — evaluated conventional and modern small area estimation approaches for aboveground forest biomass using FIA inventory data and remote-sensing-derived auxiliary variables; results presented at four national and regional research meetings.
- **Timberland Harvest Scheduling & Revenue Optimization** — evaluated a commercial forest tract under alternative management constraints and built a harvest schedule to maximize revenue, applying timber values, planning-period constraints, linear programming, NPV, and LEV to compare management strategies.
- **Python GIS Toolbox Development** — built ArcGIS Pro toolboxes integrating ArcPy, Spatial Analyst, and 3D Analyst to standardize utility GIS tasks, with documentation and user guides for team adoption.
- **Drone-Placement Suitability & Optimization** — integrated FAA, FEMA, environmental, and parcel data into a multi-criteria spatial model using cost-accumulation and spatial-overlap analysis.

EDUCATION

Master of Science — Forest Biometrics **Dec 2025**

Mississippi State University · Starkville, MS · GPA 4.0

- Thesis: *Small Area Estimation Methods for County-Level Forest Biomass Estimation.*
- Relevant coursework: Advanced Forest Inventory · Forest Biometrics · Forest Resources Management · GIS Programming · Geodatabase Design · Statistical Computations in R · Machine Learning in Life Sciences.

Bachelor of Science — Forestry **Oct 2022**

Institute of Forestry, Tribhuvan University · Pokhara, Nepal · Aggregate 80.80%

- Relevant coursework: Forest Resources Economics · Forest Mensuration · Forest Statistics · Experimental Design · GIS & Remote Sensing.

PRESENTATIONS

- County-Level Biomass Estimation Across Four FIA Units. *Partnership for Small Area Estimation Meeting*, 2025.
- Enhanced Biomass Estimation at Smaller Domains: Conventional vs. Modern Small Area Estimation Techniques. *Biannual Southern Silviculture Research Conference*, 2025.
- County-Level Aboveground Forest Biomass Estimation Using Remote-Sensing-Derived Auxiliary Information. *Graduate Student Research Symposium*, 2025.
- Machine Learning Approaches to Enhance Small Domain Estimates. *FIA Science Symposium*, 2024.

LEADERSHIP

Secretary / Coordinator **May 2019 – May 2022**

Bat Friends Pokhara · Nepal

- Led an 11-member undergraduate team in bat research and monitoring; coordinated conservation and outreach programs across five districts.

TECHNICAL PROFICIENCIES

Analysis: Python · R · SQL · pandas · NumPy · statistical modeling · machine learning

Forestry / GIS: ArcGIS Pro · QGIS · Google Earth Engine · ArcPy · lidR · terra · LiDAR · remote sensing · spatial analysis

Data & Infrastructure: PostgreSQL/PostGIS · SQLite / R-tree · geodatabases · GDAL · PDAL · FastAPI · R Shiny · Git/GitHub

TRAINING & CERTIFICATIONS

- Species Distribution Modeling with Remote Sensing — NASA ARSET webinar.
- Satellite Observations and Tools for Fire Risk, Detection, and Analysis — NASA ARSET webinar.
- Fundamentals of GIS — University of California, Davis (Coursera).
- Python Basics — Michigan State University (Coursera).
- Spatial Data Science: The New Frontier in Analytics — Esri MOOC.
- Going Places with Spatial Analysis — Esri MOOC.
- Wildlife Research Techniques Training, 2020 (19 days) — Friends of Nature (FON), Nepal.

HONORS

- Graduate Student Award, Xi Sigma Pi — Alpha Theta Chapter, 2025.
- Tribhuvan University Full Tuition Waiver, 2018–2022.